



Center for Technology Entrepreneurship
University of Washington Business School

**First West Coast Research Symposium
on Technology Entrepreneurship**

September 25-26th, 2003

AGENDA

DAY 1

Thursday, September 25th, 2003

Location: Parrington 108

8:00 to 8:30 AM: Breakfast

8:30: Welcome and introduction

8:45 to 10:00: Entrepreneurship Research: Trends and Future Directions

- S. Venkataraman (Professor, University of Virginia)
An Editor's View: What gets sent to the Journal of Business Venturing and what gets published?

10:00 to 10:30 AM: Coffee Break

10:30 to 12:15 PM: Networks, Firm Creation and Returns

- Alan Meyer (Professor, University of Oregon)

Katheryn Aten (PhD Student, University of Oregon)
Studying Organizations Far From Equilibrium: Non-Linear Change in Organizational Fields; Social Networks in Emerging Industries

- Bill Bradford (Professor, University of Washington)
Venture Capital Investment in Minority Business
- Mina Yoo (Assistant Professor, University of Washington)
The Ties That (Un)Bind: Social Networks And Entrepreneurship In High Technology Industries

12:15 to 1:15 PM: Lunch –Faculty Club

1:15 to 3:00 PM: Learning, Innovation and Entrepreneurship

- Oana Branzei (PhD Student, University of British Columbia)
Collaboration Portfolio Designs for Small Manufacturing Firms: Impacts on Absorptive Capacity and Innovation Performance
- Ilan Vertinsky (Professor, University of British Columbia)
Outcomes of Small Firms Generic and Specific Innovation Strategies: Paths to Exploration and Exploitation
- Anu Wadhwa (PhD Student, University of Washington)
Corporate Venture Capital's Impact on Rate and Direction of Innovation in Corporate Investors

3:00 to 3:15 PM Coffee Break

3:15 to 4:30 PM: Entrepreneurship through Effectuation

- Rob Wiltbank (PhD Student, University of Washington)
Effectuation And Returns To Informal Venture Capital Investing
- Saras Sarasvathy (Assistant Professor, University of Maryland)
Suresh Kotha (Professor, University of Washington)
Dealing with Knightian uncertainty in the new economy: The Real Networks case
- Discussion and wrap up

DAY 2

Friday, September 26th, 2003

Location: Parrington 108

8:00 to 8:30 AM: Breakfast

8:30 to 10:00 AM: Technology, Innovation, and Entrepreneurship

- Corey Phelps (Assistant Professor, University of Washington)
Technological Exploration: A Longitudinal Study of the Role of Technical Diversity and Social Capital in Alliance Networks
- Sanjay Jain (Assistant Professor, University of Wisconsin-Madison)
From ape to human: Explicating the processes involved in compatibility standards evolution
- Raghu Garud (Professor, New York University)
Breaking Through and Breaking Apart: Microdynamics of Technological Revolutions

10:00 to 10:30 AM: Coffee Break

10:30 to 12:00 PM: New Venture Creation and Survival

- MitraBarun Sarkar (Assistant Professor, University of Central Florida)
Knowledge Transfer Through Inheritance: Spin-out Generation, Development and Survival
- Aviad Pe'er (PhD Student, University of British Columbia)
Enterprise Failures as a Determinant of Entrepreneurial Entry: A Study of Within Region

Micro-Clusters

- Harry Sapienza (Professor, University of Minnesota)
Capability Development in New Ventures: The Implications of Early Internationalization on Survival and Growth
- Wrap up Discussion

Lunch (Parrington Commons) and Adjourn



Center for Technology Entrepreneurship
University of Washington Business School

First West Coast Research Symposium
on Technology Entrepreneurship
September 25-26th

Presentation Titles and Abstracts

Katheryn Aten, PhD Student, University of Oregon
Social Networks in Emerging Industries

Kathryn will present the conceptual core of her dissertation proposal. This study is part of the larger research program that Professor Alan Meyer will be summarizing.

Bill Bradford, Professor, University of Washington
Venture Capital Investment in Minority Business

This paper analyzes the investments and financing of venture capital (VC) firms that focus on investing in minority-owned businesses (MBEs). Minority oriented VC funds do not concentrate in high tech industries; they support a more diverse range of industries. The minority-focused funds vary enormously in size: seven of the 24 funds responding to our survey started with less than \$10 million, and five began with over \$50 million. In order to gain diversification and offset limited managerial staff, a universal strategy of the funds is participating in syndicated MBE investments. Public pension funds have been an important source of funds: their median investment in a minority VC fund is \$55 million. Because industry newcomers are not able to

tap pension funds or funds of funds, the bank/insurance company group is critically important to the industry's long-term development: their funding is accessible to new industry entrants. Minority-focused VC funds have earned yields that are comparable with those of the broader VC industry, despite the restriction of investing in minority businesses. This implies that the negative bias against minorities previously found in VC markets (Bates and Bradford, 1992) creates arbitrage opportunities for minority focused VC funds.

Oana Branzei, PhD Student, University of British Columbia
Ilan Vertinsky, Professor, University of British Columbia
Collaboration Portfolio Designs for Small Manufacturing Firms: Impacts on Absorptive Capacity and Innovation Performance

This study examines how the characteristics of small firms' collaboration portfolios influence their learning and innovation. The paper investigates the separate effects of partner-type heterogeneity (collaborating with diverse types of partners) and geographic-region heterogeneity (collaborating with partners from different regions) on two sets of dependent variables. It addresses two related research questions: (1) Can more heterogeneous collaborations help small firms acquire, assimilate, transform, and exploit knowledge from external sources? (2) Can more heterogeneous collaboration portfolios improve the pace, speed, novelty, effectiveness, or commercialization of small firms' innovation? The paper discusses a paradox in organizational learning. On the one hand, firms have most to learn from dissimilar others – organizations with different specialities and mental models (Cohen & Levinthal, 1990; Gavetti & Levinthal, 2000), organizations from different industries (Kessler, Bierly & Gopalakrishnan, 2000; Rosenkpf & Nerkar, 2001), or organizations embedded in different clusters of capabilities (Frost, 2001). On the other hand, it is very difficult to learn from a partner unless the knowledge bases, structures,

and priorities of the two organizations provide some overlap that facilitates exchanges and joint development of capabilities (Lane & Lubatkin, 1998). The study joins the mainstream of the learning literature, arguing that direct involvement in joint projects can foster path-breaking learning (Huber, 1991; Walsh, 1995) which then boosts firms' capability to innovate. When firms associate themselves with different partners they gain access to different "pockets of capabilities" and they may be inspired to jointly create radically new knowledge. Current studies on alliances and learning have offered some conflicting results. Powell, Koput & Doerr (1996) argue that greater involvement in diverse networks facilitates innovation. Yet Ha (2001) finds only certain types of organizations learn from their alliances, and the benefits of alliances vary by industry and firm life cycle (Bird Schoonhoven & Lyman, 2000). In this study we focus on small manufacturing firms, and test the direct relationships between the characteristics of their alliance portfolios, different components of absorptive capacity, and several types of innovation outcomes.

Outcomes of Small Firms Generic and Specific Innovation Strategies: Paths to Exploration and Exploitation

This study discusses what to what extent firms' strategies as well as their strategic positioning relative to competitors within the same industry can offer small entrepreneurial firms a competitive advantage. We focus on the impact of small firms' *generic and specific innovation strategies* on three types of innovation outcomes – the ratio of exploration to exploitation, the pace of innovation, and the number of patent applications. *Generic innovation strategies* include the adoption a pioneering stance, the emphasis placed on internal R&D, the importance of external collaborations, the scope of product and process innovation, and the scope of the market

a firm is after. *Specific innovation strategies* refer to a) recombination, i.e. firms' internal efforts to update their production processes by seeking and combining prior skills with new technologies, and b) the novelty of the inputs used in the production process. We are interested in the direct effects of generic and specific innovation strategies on small firms' innovation performance. We also investigate whether specific innovation strategies mediate the influence of generic strategies on innovation outcomes.

Raghu Garud, Professor, New York University
Breaking Through and Breaking Apart: Microdynamics of Technological Revolutions
(with Kamal Munir)

This paper explores the dynamics of technological breakthroughs in socio-technical ensembles. The partition of these ensembles into modules reduces cognitive complexity for associated actors. At the same time, these modules are embedded in larger social arrangements that embody the interests of the stakeholders involved. Breakthroughs set in motion a sequence of events that can end up destroying these arrangements. We explore these dynamics in the case of Polaroid's landmark design, the SX-70 camera. The creation of this design required a recombination of knowledge contained across once modular components. However, with recombination, the interests of the actors embedded in the modules were compromised. As a result, the integration of camera modules led to the disintegration of surrounding social arrangements required to support the wider diffusion of the SX-70 camera.

Sanjay Jain, Assistant Professor, University of Wisconsin-Madison

From ape to human: Explicating the processes involved in compatibility standards evolution

Compatibility standards refer to the interface specifications that provide users and vendors with a common platform and ensure connectivity between products. While the emergence of such standards can contribute to the widespread evolution of a technology, over time they constrain a technology from evolving to new functionalities. Given this scenario, I ask the question: ***how do compatibility standards evolve?*** In this paper, I suggest that compatibility standards that are assumed to be locked-in in fact evolve considerably over time. More importantly, the development of extensions to the original specification is often critical to the long-term survival prospects of the compatibility standard. I refer to such specifications as ***extensible standards*** and examine the processes by which extensions to the compatibility standards are created, the role of agents (and the nature of agency) in the creation of such extensions and the manner in which agents maintain continuity with the existing standard even as they extend it.

Alan Meyer, Professor, University of Oregon
Studying Organizations Far From Equilibrium: Non-Linear Change in Organizational Fields

The thrust of the presentation will be a summary of a multi-investigator NSF-funded research program that we're pursuing at the UO. The central focus is industry emergence and discontinuous change in organizational fields. The empirical data come from corporate venturing programs and a network analysis of the nanotechnology investing community.

Aviad Pe'er, PhD Student, University of British Columbia
Ilan Vertinsky, Professor, University of British Columbia

Enterprise Failures as a Determinant of Entrepreneurial Entry: A Study of Within Region Micro-Clusters

The dynamics of entry and failure determines the growth and sustainability of clusters. Both Organizational Ecology and Industrial Organization literatures predict the existence of positive correlation between entry and failure rates. This relationship is attributed mainly to the liability of newness, links between entry and failure barriers and adverse selection. In this paper we hypothesize alternative causal links between enterprise failure and entrepreneurial entry. We posit that the casual links also flow from failure to entry. Failure levels affect both the intensity of entrepreneurial search for opportunities and the opportunity landscape facing entrepreneurs. Failure rates provide a signal about the difficulties of succeeding in a particular location. Controlling for agglomeration economies, competition, diversity of the economic environment, economic region, industry and time fixed effects, we found that failures in a location and neighboring locations are positively associated with entry levels. Higher failure rates in a location, however, deter entry, while higher failure rates in neighboring locations increase its attraction to new entrants. Our study also shows that agglomeration economies and failure level effects attenuate with distance. The model is tested using a unique longitudinal establishment-level panel database made available by Statistics Canada, which encompasses all Canadian enterprises between 1984 and 1998.

Corey Phelps, Assistant Professor, University of Washington
Technological Exploration: A Longitudinal Study of the Role of Technical Diversity and Social Capital in Alliance Networks

Strategic renewal and innovation are central elements in the process of corporate entrepreneurship. In this study I examine how the strategic alliances in which established firms participate affect a fundamental mechanism of renewal and innovation: technological exploration (i.e., the creation of relatively novel technological knowledge). Drawing on the concept of

recombinatory search I argue that access to technically diverse alliance partners provides the benefits of variety and an enlarged search space, leading to a positive effect on technological exploration. Second, building upon social capital theory I posit that the extent to which a firm's alliance partners are densely interconnected will generate trust and reciprocity among partners, increasing their cooperation in sharing knowledge. I predict that these benefits (or *structural* social capital), and thus network density, will have a positive effect on technological exploration. Longitudinal data on a panel of 57 global telecommunications equipment manufacturers covering the period 1987-1997 are used to test these predictions. Patent data are used to measure the technological dissimilarity of alliance partners and firm-level technological exploration. The alliance data set was constructed from an extensive archival search of over 160,000 documents. Taken together, the findings of the study provide evidence that alliances do indeed effect firm exploration. Both technological diversity and alliance network density had a positive effect on technological exploration. Furthermore, network density enhanced the positive effect of diversity on technological exploration.

Harry Sapienza, Professor, University of Minnesota
Capability Development in New Ventures: The Implications of Early Internationalization on Survival and Growth
(with Erkkö Autio, Gerry George, and Shaker Zahra)

In this article, we propose a model on early internationalization and its effects on the survival and growth prospects of young firms. We argue that, *ceteris paribus*, earlier internationalization decreases survival prospects while increasing growth prospects. We derive a capability-based view of internationalization and extend theory by detailing how the firm's age at internationalization, its absorptive capacity, and its slack resources moderate the effects of

internationalization. The article contributes to ongoing dialog in the dynamic capabilities and the international entrepreneurship literatures.

Saras Sarasvathy, Assistant Professor, University of Maryland
Dealing with Knightian Uncertainty in the New Economy: The Real Networks Case
(with Suresh Kotha)

Using a revelatory case study of the creation of RealNetworks, a leading internet firm specializing in streaming media, we document the existence of three key hypothesized patterns of events based on causation and effectuation in entrepreneurship. Specifically, we demonstrate how RealNetworks used the three principles of effectuation embedded within the logic of *non-predictive* control – (1) affordable loss, (2) strategic partnerships, and (3) leveraging contingencies – to deal with complex and multiple manifestations of Knightian uncertainties in its micro and macro decision environments.

MitraBarun Sarkar, Assistant Professor, University of Central Florida
Knowledge Transfer Through Inheritance: Spin-out Generation, Development and Survival
(with Rajshree Agarwal, Raj Echambadi, and April Franco)

This paper examines the role of inherited knowledge in an organization's formation and as a subsequent source of its competitive advantage. We investigate how knowledge inherited from an industry incumbent by a "spin-out" (an entrepreneurial venture by an ex-employee) influences the spin-out's development and performance. Using data from the disk drive industry from 1977 to 1997, we show that the incumbent's capabilities related to technology and market pioneering predict spin-out formation by its employees. Results show that incumbents with both strong technological *and* market pioneering know-how generate fewer spin-outs than firms with strengths in only one area. The incumbent's capabilities at the time of spin-out founding

positively affects the spin-out's knowledge capabilities and result in spin-outs having higher probabilities of survival relative to other industry entrants.

S. Venkataraman, Professor, University of Virginia

An Editor's View: What gets sent to the Journal of Business Venturing and what gets published?

Venkat, based on analysis of the papers sent to JBV over the last 4-5 years, will raise some provocative questions on what are the really important questions in entrepreneurship and what gets addressed in the journal. In doing this, he will set the stage for a lively discussion of the current state of entrepreneurship research.

Anu Wadhwa, Doctoral Candidate, University of Washington

Corporate Venture Capital's Impact on Rate and Direction of Innovation in Corporate Investors

Firm level theories of innovation and technological change suggest that when firms increase their knowledge bases, their innovativeness increases. Prior literature has studied various drivers of innovation such as patent search behavior, alliances, and acquisitions. However, despite the increasing use of corporate venture capital investments by large, established firms in private entrepreneurial companies as a tool to gain strategic benefits, the link between corporate venture capital investments and innovation search behavior in the corporate investor has not been examined. I explore whether corporate venture capital investments impact innovation in the corporate investor. Using evolutionary perspectives and learning and knowledge based theories of the firm, I propose that corporate venture capital investments stimulate innovation in the corporate investor. I further argue that the attributes of the corporate investor and characteristics of the corporate investor-portfolio firm relationship determine the efficacy of corporate venture

capital as a mechanism to increase innovation. I also examine whether corporate venture capital investments in certain firms enhance exploratory innovation in the corporate investor. This dissertation proposal makes both theoretical and empirical contributions to the literatures on innovation search, knowledge based view of the firm and corporate entrepreneurship.

Rob Wiltbank, PhD Student, University of Washington
Effectuation And Returns To Informal Venture Capital Investing
(with Saras Sarasvathy)

In spite of the fact that informal venture capital, Angel investment, represents the largest source of private equity capital in new venture development, from a research standpoint we know extremely little about this type of investment. The bulk of what we do know results from mainly from investigations into formal venture capital, which, in turn, leans heavily on financial theories from larger scale traditional equity markets. Angel investing, however, takes place in settings of much greater uncertainty, and through a process that is often very different than that of formal venture capitalists. Therefore, the role of agency risks, predicted net present values of future cash flows, overcoming contractual hazards due to opportunism, information asymmetry, etc. may well be overstated. In this paper, we propose that while venture capital research provides a useful starting point for research into angel investment, the decision processes used by angels and VCs differ significantly in their cognitive content and overall logic, and that this difference will impact the structure and returns to the two types of investing. While VCs tend to use predominately causal approaches based on predictive rationality, angels are likely to be more effectual and make their choices based on the logic of non-predictive control. By clarifying key characteristics of current theorizing and practice in private equity investing, and detailing

additional perspectives from the theory of effectuation, we hope to motivate and inform additional empirical research in this critical area in entrepreneurship research.

Mina Yoo, Assistant Professor, University of Washington

The Ties That (Un)Bind: Social Networks And Entrepreneurship In High Technology Industries

Together with human capital and financial capital, social capital is seen as completing the triumvirate of resources required for individual and organizational achievement, including in the context of entrepreneurship. I examine the ways in which social networks are used by entrepreneurs and organizations at distinct stages of their evolution – at the early stages of venture creation and as public corporations – and the effects that the networks have on performance at these stages. This research consists of three studies using narratives, surveys, and archival data. In the first study, I conduct in-depth interviews with 45 entrepreneurs and find that entrepreneurs select their relationships based on the level of uncertainty surrounding their ventures at different stages of development and their perceived accessibility of the relationships at these stages. In the second study, I use propositions derived from the first study to develop a model predicting 1) the effect of entrepreneurial stage (the start-up and the expansion stages) on network characteristics, consisting of network density (level of connectedness among members of the network), heterogeneity, and co-ethnicity (extent to which the members of the networks share the same ethnicity as the entrepreneur) and 2) the effect of these network characteristics on performance, conceptualized as amount of funding obtained at each stage. I then test the model using data collected through a survey of 91 Indian and Chinese immigrant entrepreneurs and 60 native-born white American entrepreneurs, highlighting differences between these groups. I find that the network characteristics as well as their effect vary by entrepreneurial stage and by

immigrant/ethnic status. In the third and final study, using a new sample consisting of 362 public corporations in Silicon Valley, I predict organizational failure during the stock market crash of the year 2000. I find that the level of connectedness among the members of the organizational networks have an inverse U-shaped relationship with the propensity to fail. By distinguishing between entrepreneurial stages as well as entrepreneur characteristics and by investigating the effects as well as the formation of networks along multiple dimensions, my studies collectively extend the literature on social networks, entrepreneurship, and immigrant business activity.